

Linux Users and Groups

1. Introduction

- Linux is a multi-user operating system, meaning multiple human users and system processes (daemons) can operate on the same system simultaneously.
 - Users: Entities (human or process) that can interact with the OS.
 - Groups: Collections of users used for easier permission management.
 - This structure enhances security, access control, and organization.
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2. Users in Linux

What a User Account Contains

Each user account is uniquely identified by:

- Username → Human-readable name.
 - UID (User ID) → Numerical ID for the user (used internally by the system).
 - Home Directory → Private file storage space.
 - Shell → Default command-line environment.
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Types of Users

1. Root User

- UID = 0.
- Full system access and permissions.
- Can read/modify system files, install/remove software, configure hardware, and start/stop services.
- Used for administration tasks — *should not be used for daily work*.

2. Regular User

- UID range: 1000+ (varies by distro).
- Created for human users to perform daily, non-critical tasks.
- Limited to their home directory and given files.
- Requires sudo to run admin commands.

3. System User (*Expanded Explanation*)

- UID range: 1–999 (or similar, depending on distro).
- Represents background services rather than humans (e.g., `mysql`, `www-data`, `postfix`).
- No login shell (`/usr/sbin/nologin` or `/bin/false`).
- Purpose:
 - Run services securely with limited privileges.
 - Isolate service processes for security (if one is compromised, others remain safe).
 - Manage access to files related to that service only.
- Example Use Cases:
 - `mysql` → owns database files and processes.
 - `www-data` → runs Apache/Nginx web server processes.
 - `sshd` → manages SSH remote sessions.

3. Where User Information is Stored

- `/etc/passwd`
 - Stores account details (username, UID, GID, home, shell, comment field).
 - Password field is typically `x` (indicating it's stored in `/etc/shadow`).
 - Example:

```
jack:x:1001:1001:Akshay:/home/jack:/bin/bash
```

- Breakdown:
 - `jack` → username
 - `x` → password in `/etc/shadow`
 - `1001` → UID
 - `1001` → primary GID
 - `jack` → comment/full name
 - `/home/jack` → home directory
 - `/bin/bash` → default shell

- /etc/shadow
 - Stores encrypted passwords and password aging.
 - Example:

```
jack:$6$hashvalue:19236:0:99999:7:::
```

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4. Groups in Linux

- Definition: A group is a collection of users used to manage permissions collectively.
- Each group is identified by:
 - Group Name
 - GID (Group ID)
 - Members (usernames)

Types of Groups

1. Primary Group
 - Assigned when a user is created.
 - When the user creates a file, it is owned by their primary group.
2. Secondary Groups
 - Additional groups the user is a member of for accessing shared resources.
 - A user can belong to multiple secondary groups.

Where Group Information is Stored

- /etc/group → Stores group info.
- /etc/gshadow → Stores encrypted group passwords and admin settings.

Example from [/etc/group](#):

```
developers:x:1002:akash,vaibhav
```

5. User Management Commands

Task	Command Example
Add User	<code>useradd username</code>
Add User + Password	<code>useradd jack → passwd jack</code>
View Users	<code>cat /etc/passwd</code>
Modify User	<code>usermod</code>
Add to Secondary Group	<code>usermod -aG groupname username</code>
Change Shell	<code>usermod -s /bin/zsh username</code>
Delete User	<code>userdel username</code>
Delete User + Home Dir	<code>userdel -r username</code>

6. Group Management Commands

Task	Command Example
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Add Group	<code>groupadd developers</code>
View Groups	<code>cat /etc/group</code>
Add User to Group	<code>usermod -aG developers vaibhav</code>
Change Group Name	<code>groupmod -n newname oldname</code>
Delete Group	<code>groupdel developers</code>
Set Group Admin	<code>gpasswd -A username groupname</code>
Switch to Group	<code>newgrp groupname</code>

7. Advanced User & Group Management

- Custom Home Directory:
`useradd -d /opt/users/john john`
- Account Expiry Date:
`useradd -e 2024-12-31 john`
- Assign Custom UID or GID:
 - `useradd -u 2001 username`
 - `groupadd -g 1050 groupname`
- Lock/Unlock Accounts:
 - Lock: `passwd -l username`
 - Unlock: `passwd -u username`

8. UID (User ID) in Linux

- Definition: Unique numerical ID for a user. The system uses UID internally instead of username.
 - UID Ranges:
 - 0 → root
 - 1–99 → reserved for predefined system accounts
 - 100–999 → system/service accounts (non-human)
 - 1000+ → regular human users
 - Use in System:
 - File ownership
 - Process execution permissions
 - Access control
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9. How UID and GID Work Together

- UID → identifies the user.
 - GID → identifies the primary group.
 - Together → determine ownership and permissions.
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10. Best Practices

- Use groups to manage permissions instead of per-user changes.
 - Avoid using root account daily; instead use `sudo`.
 - Assign system users for services rather than running them as root.
 - Regularly audit `/etc/passwd` and `/etc/group` for unused accounts.
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11. Practice Questions

1. Command to create new user `john`.
2. Where is Linux user account info stored?
3. Difference between primary and secondary group?
4. Command to add `john` to `developers` group.
5. Purpose of `/etc/shadow`.
6. Create user `mike` with home at `/opt/mike`.
7. Lock/unlock user `alan`.

8. Purpose of `gpaswd`.
 9. UID difference between 0 and 1000.
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